

Mapping AI diagnostic innovation networks in sub-Saharan Africa: a social network analysis of trials, institutions, and funders



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Summary

Background Artificial intelligence (AI) diagnostic technologies are emerging tools for disease screening in sub-Saharan Africa, yet the partnership structures shaping their clinical validation remain uncharacterised. We conducted an exploratory network analysis to map the relational structure of AI diagnostic clinical trial networks across the region.

Methods We conducted a social network analysis of all registered AI diagnostic clinical trials in sub-Saharan Africa through November 30, 2024, identified via systematic searches of [ClinicalTrials.gov](https://clinicaltrials.gov), PACTR, and the WHO ICTRP. We extracted institutional affiliations, funding sources, and technology partnerships from trial registries and characterised network structure using degree, betweenness, and closeness centrality.

Findings We identified 11 registered trials across 10 countries involving 35 institutions (46 nodes, 97 edges; density 0.094). Trials concentrated in East Africa (36.4%) and addressed tuberculosis (27.3%), cervical cancer (27.3%), and diabetic retinopathy (18.2%). Six trials (54.5%) had completed enrolment; 72.7% were registered post-2020. Funding was diversified across US government agencies, private foundations, and Swiss institutions. Network analysis revealed 11 disconnected components with uniformly zero betweenness centrality, indicating no institution occupied a bridging position between trials. Degree centrality ranged 0.022–0.178 (mean 0.094 ± 0.051).

Interpretation AI diagnostic clinical trial networks in sub-Saharan Africa exhibit complete structural fragmentation with no institutional bridging positions, suggesting limited cross-trial knowledge transfer despite active participation. This pattern likely reflects nascent field development, disease-specific siloing, and project-based funding. Structural connectivity alone cannot determine ecosystem sustainability — genuine capacity building requires regulatory approvals, local research leadership, and health system integration that network metrics cannot assess.

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Introduction

Across sub-Saharan Africa, persistent diagnostic gaps hinder timely detection and management of priority conditions including tuberculosis, HIV, malaria, maternal and neonatal complications, and non-communicable diseases.^{1,2} Artificial intelligence (AI)-

enabled diagnostics, spanning computer vision for imaging, signal interpretation, and data-driven triage, are increasingly tested to mitigate workforce and infrastructure constraints and to improve reach and quality of services.^{3–5} Investments by governments, multilaterals, philanthropy, and industry have produced

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Research in context

Evidence before this study

We searched PubMed, Web of Science, and Google Scholar for systematic reviews and network analyses of AI diagnostic applications in sub-Saharan Africa published through November 30, 2025, using terms including “artificial intelligence,” “machine learning,” “diagnostics,” “Africa,” and “clinical trial network.” Prior research documented individual pilot implementations and algorithm development but did not systematically characterise clinical research networks or partnership structures. Systematic reviews of AI for health in low- and middle-income countries identified persistent challenges around data governance, regulatory readiness, and equity, but none mapped collaboration networks across trials. Network analyses of other health innovation domains in sub-Saharan Africa revealed Northern institutional dominance; no studies had mapped AI diagnostic trial networks specifically.

Added value of this study

This study provides the first network analysis of AI diagnostic clinical trials in sub-Saharan Africa. Among 11 registered trials across 10 countries involving 35 institutions, network analysis revealed complete fragmentation into isolated trial-specific components with uniformly zero betweenness centrality. No institution occupied a bridging position

facilitating knowledge transfer between trials. Geographic patterns showed East African concentration, with Lesotho hosting the most trials ($n = 2$) driven by tuberculosis research. Funding proved more diversified than expected, spanning US government agencies, private foundations, and Swiss institutions rather than single-funder dominance.

Implications of all the available evidence

Current AI diagnostic trial networks exhibit structural fragmentation that may constrain knowledge accumulation despite active institutional participation. The absence of bridging positions means methodological innovations, regulatory navigation strategies, and community engagement lessons developed in one trial are unlikely to diffuse to others through documented institutional channels. However, structural connectivity alone cannot determine ecosystem sustainability — genuine capacity building requires regulatory approvals, local research leadership, and health system integration that network metrics cannot assess. Interventions should create cross-trial learning mechanisms while addressing power dynamics through governance arrangements that ensure African institutional leadership, equitable data ownership, and capacity that persists beyond individual trial lifecycles.

pilots and early trials across chest radiography for tuberculosis, obstetric ultrasound, cervical cancer screening with digital pathology, dermatology, and point-of-care testing. Yet the evidence base remains fragmented and heterogeneous with relatively few prospective evaluations of diagnostic innovations for the region. Innovation network reports that do exist highlight concerns regarding transparency and reproducibility, regulatory readiness, and equity in data governance, authorship, and benefit sharing.^{6,7}

Innovation ecosystems and structural foundations

Innovation and evidence generation in AI diagnostics are shaped by collaboration patterns, including who partners with whom, where funding flows, and how knowledge travels. An innovation ecosystem, in its broadest sense, mobilizes diverse actors, assets, and relationships to provide a supportive enabling environment for innovation to flourish.^{Jackson, 2011 #42787} These ecosystems are inherently dynamic, comprising an ever-changing array of actors and institutions with shifting patterns of power, resources, and relationships. Ecosystem boundaries are rarely clear-cut, with actors, assets, and relationships often spread across multiple levels and geographies—from local clinical sites to regional research networks to international funding organizations.⁸

The range of actors forming innovation ecosystems typically includes: government agencies and regulators, private sector companies, research institutions and universities, clinical trial sites, development agencies, venture capital and philanthropic funders, as well as intermediaries who facilitate the transfer of ideas, technology, and resources.⁸ Understanding who participates in innovation networks, how they connect, and what patterns of collaboration emerge is critical for building research capacity and ensuring equitable access to transformative health technologies in resource-limited settings.

This analysis focuses specifically on AI diagnostic technologies as one component of broader AI health innovation in sub-Saharan Africa. AI diagnostics represent an important but limited domain—other applications include treatment optimization, resource allocation, disease surveillance, and health system management. We focus on diagnostics for three reasons. First, diagnostic tools typically require clinical trial validation before deployment, generating documented institutional networks in trial registries that enable systematic network analysis. Second, many high-burden conditions in sub-Saharan Africa (tuberculosis, cervical cancer, diabetic retinopathy) have established diagnostic gaps where AI shows promise for improving access and accuracy. Third, computer

vision and machine learning for diagnostic image interpretation represent relatively mature AI applications compared to emerging therapeutic or health systems tools. This diagnostic focus enables systematic analysis through trial registries but necessarily excludes AI innovations in treatment, prevention, and implementation science that may follow different development and regulatory pathways and partnership structures.

Network analysis as an ecosystem characterization tool

Social network analysis (SNA) offers tools to characterize research and trial ecosystems by modeling institutions, countries, and funders as nodes, and co-authorships, trial partnerships, and funding ties as edges. In health systems research, SNA has been widely used to help improve the effectiveness and efficiency of decision-making processes,⁹ particularly in understanding how stakeholders interact within complex and dynamic systems.¹⁰ The method provides an appropriate and innovative paradigm for health systems researchers, allowing new analyses of system structure and facilitating understanding of stakeholder roles.¹¹

The network perspective is particularly useful for understanding, monitoring, or evaluating implementation processes when programs, policies, or principles are designed and scaled up or adapted to different settings.¹² In health research contexts, SNA has exposed gatekeeping and collaboration bottlenecks¹³ and highlighted opportunities for targeted capacity strengthening and improved knowledge diffusion.¹⁴ However, SNA remains underused in characterizing innovation ecosystems in low- and middle-income country health systems.¹⁰

We therefore conducted an exploratory ecosystem network analysis to map the relational structure underlying AI diagnostic research and trials in sub-Saharan Africa. This study represents an applied use of established SNA methods to characterize the emergent structure of AI diagnostic clinical trial networks, a context where these approaches have not previously been used. Our approach complements traditional systematic reviews of trial efficacy by revealing structural patterns and relationship configurations that influence innovation diffusion and ecosystem development.

Methods

Definition of AI diagnostics clinical trials

We use “clinical trial” as umbrella term encompassing both diagnostic validation studies (assessing AI test accuracy against reference standards) and diagnostic intervention trials (assessing whether AI-guided diagnosis improves health outcomes). We included studies registered in recognized trial registries ([ClinicalTrials.gov](https://clinicaltrials.gov),

PACTR, WHO ICTRP) involving prospective human participants and systematic AI technology evaluation. This inclusive approach captures the spectrum of formal clinical research on AI diagnostics, recognizing definitional ambiguity in diagnostic test evaluation terminology. All trials involved prospective enrollment, ethical approval, and formal registration regardless of whether they met narrow therapeutic intervention trial definitions.

Data extraction and network construction

We developed a standardized extraction framework capturing multiple entity types and their relationships: (1) Clinical trials: Trial identifiers, disease focus areas, AI technologies employed, sample sizes, and outcomes; (2) Institutions: Academic institutions, hospitals, research centers, and their roles within trials; (3) Commercial entities: Technology companies, diagnostic manufacturers, and their specific contributions; (4) Funding organizations: Government agencies, foundations, and private funders; (5) Geographic information: Countries, regions, and clinical sites; (6) Temporal data: Trial registration dates, completion timelines, and funding periods. Network relationships were defined as direct connections between entities participating in the same trial, funding relationships, technology transfer arrangements, and institutional collaborations documented in trial registrations or related publications.

Network construction, assumptions, and edge definitions

We constructed a bipartite network from trial registry and publication data following established social network analysis approaches in health systems research.^{10,11,15} Our network includes two node types: (1) trials and (2) institutions. Institutions are categorized by sector attribute into academic institutions (54.3%, 19/35), funding organizations (31.4%, 11/35), technology companies (11.4%, 4/35), and government entities (2.9%, 1/35). We treated all registered trials meeting inclusion criteria as a complete census of AI diagnostic clinical trial activity in sub-Saharan Africa rather than a statistical sample, following established practice for whole-network case studies.

Direct trial-institution edges were created when institutions were listed as principal investigators, collaborating sites, or implementing partners in trial registrations. Co-participation edges (institution-institution connections) were constructed when two institutions participated in the same trial, creating an institutional collaboration network projected from the trial-institution bipartite graph. Funding edges connected funders to trials and institutions they supported. Technology provision edges linked companies to trials using their AI systems. All edges were treated as unweighted, representing the presence or absence of

relationships rather than strength or intensity of collaboration.

Network analytics

The network was developed using Python 3.12 with NetworkX for graph construction and analysis.⁷ We present network metrics as descriptive characterizations of observed relational structure rather than inferential statistics, following guidance for non-independent network data.¹¹

Centrality measures included degree centrality (normalized proportion of possible connections) to identify well-connected institutions, betweenness centrality (proportion of shortest paths passing through a node) to identify potential brokers between otherwise disconnected network components, and closeness centrality (inverse average distance to all other nodes) to assess accessibility within the network. Network-level metrics included density (proportion of actual to possible edges), average clustering coefficient (tendency for connected nodes to share neighbors), and number of connected components (distinct subgraphs with no connections between them). Robustness analysis examined how network structure changed with systematic removal of high-degree nodes to assess dependence on specific hub institutions ([Supplementary Table S4](#)). Subnetwork analyses focused on country-specific innovation networks, disease-focused clusters, and funding relationship patterns.

Complete methodological documentation including search strategies with registry-specific syntax, data extraction protocols, network construction decision rules, quality assurance procedures, and Living Systematic Review Platform governance is provided in [Supplementary Materials](#) to ensure full reproducibility.

Results

Trial characteristics and geographic distribution

Our comprehensive search identified 11 registered AI diagnostic clinical trials conducted in sub-Saharan Africa, spanning 10 countries across the region ([Table 1](#)). Lesotho hosted two trials, while Botswana, Cameroon, Kenya, Mozambique, Nigeria, Rwanda, Tanzania, Zambia, and Zimbabwe each hosted one trial. The majority of trials (72.7%, 8/11) were registered on [ClinicalTrials.gov](#), with two trials registered in the Pan African Clinical Trials Registry (PACTR) and one in the ISRCTN registry.

Trials addressed diverse diagnostic challenges, with tuberculosis and cervical cancer screening representing the most common target conditions (27.3% each, 3/11 trials). Two trials (18.2%) focused on diabetic retinopathy screening, while individual trials addressed peripartum cardiomyopathy, primary care diagnosis using large language models, and glaucoma detection. The

median planned enrollment was 2000 participants (IQR: 1500–2100; range: 100–9000), with total enrollment across all trials reaching 47,474 participants.

Technology platforms varied across trials, with CAD4TB (Computer-Aided Detection for Tuberculosis) employed in three trials (27.3%), representing the most widely deployed AI system. Deep learning algorithms were employed in two trials (18.2%) for diabetic retinopathy screening, while smartphone-based AI classification, AI-enabled electrocardiography, large language models, fundus photography algorithms, and the Neo Sense Vector classifier were each evaluated in single trials. Most trials (63.6%, 7/11) were designed as Phase 2 validation studies, with three trials (27.3%) classified as Phase 3 pragmatic or clinical trials, and one trial (9.1%) lacking phase classification. Six trials (54.5%) had completed enrollment, two were recruiting (18.2%), one was active but not recruiting (9.1%), and two had not yet begun recruitment (18.2%).

Temporal evolution of AI diagnostic research

Trial registration activity increased substantially over the study period, with 72.7% (8/11) of trials registered in 2021 or later ([Fig. 1](#)). The earliest trial was registered in 2018 ($n = 1$), with a second registered in 2019, followed by steady growth with 1–2 new trial registrations annually through 2025. This temporal concentration reflects the nascent state of AI diagnostic clinical research in sub-Saharan Africa, with the field demonstrating rapid expansion in recent years coinciding with increased global attention to AI applications in healthcare and expanded funding for digital health innovations in resource-limited settings.

Network structure and composition

The complete AI diagnostic trial network comprised 46 nodes—11 clinical trials and 35 participating institutions—connected by 35 direct trial-institution relationships, which when expanded to include co-participation edges (institutions connected through shared trial participation) resulted in 97 total edges ([Table 2](#), [Fig. 2](#)). The network exhibited sparse connectivity, with an overall density of 0.094, indicating that only 9.4% of all possible connections between nodes were present. Average degree centrality across all nodes was 4.22, with degree centralization of 0.002, indicating a highly decentralized network structure lacking dominant central hubs.

A critical structural characteristic emerged from network component analysis: the network fragmented into 11 completely disconnected components, with each component consisting of a single trial and its associated institutions. No institutions participated in multiple trials, resulting in isolated trial-specific collaboration networks with no inter-trial connections. This structural pattern reflects a trial-centric network architecture

No.	Trial ID	Registry	Country	Target condition	AI technology	Lead institution	Registration year	Status	Planned/ Actual enrolment ^a	Results published	DOI
1	NCT03757299	ClinicalTrials.gov	Cameroon	Cervical cancer	GeneXpert HPV test + VIA/VILI	University Hospital Geneva (HUG)	2018	Completed	4473	No	NA
2	NCT04242823	ClinicalTrials.gov	Botswana	Cervical cancer	Smartphone-based AI classifier	Beth Israel Deaconess Medical Center	2020	Active, not recruiting	3000	No	NA
3	PACTR202101512465690	PACTR	Rwanda	Diabetic retinopathy	Cybersight AI retinal grading	Rwanda International Institute of Ophthalmology (RIIO)/Orbis International	2021	Completed	250 planned; 273 actual	Yes	https://doi.org/10.1016/j.xops.2022.100168
4	NCT04666311	ClinicalTrials.gov	Lesotho; South Africa	Tuberculosis	CAD4TB (Delft Imaging) + CRP assay	Swiss TPH	2020	Completed	1400 planned; 1392 actual	Yes	https://doi.org/10.1093/cid/ciae378
5	NCT05139940	ClinicalTrials.gov	Zambia	Tuberculosis	Google TB AI (CXR TB AI)	Centre for Infectious Disease Research in Zambia (CIDRZ)	2021	Completed	2432 planned; 1978 actual	Yes	https://doi.org/10.1056/Aloa2400018
6	NCT05526885	ClinicalTrials.gov	Lesotho; South Africa	Tuberculosis	CAD4TB v7 + POC-CRP	Swiss TPH	2022	Completed	20,000	No	NA
7	NCT05438576	ClinicalTrials.gov	Nigeria	Peripartum cardiomyopathy	AI-enabled ECG + digital stethoscope	Mayo Clinic	2022	Completed	1400 planned; 1232 actual	Yes	https://doi.org/10.1038/s41591-024-03243-9
8	NCT06042543	ClinicalTrials.gov	Zimbabwe	Cervical cancer	Neo Sense Vector AI classifier	IARC-WHO	2023	Recruiting	3200	No	NA
9	ISRCTN18317152	ISRCTN	Tanzania	Diabetic retinopathy	Deep learning AI retinal grading	London School of Hygiene & Tropical Medicine	2023	Not Recruiting	2364	No	NA
10	NCT06552247	ClinicalTrials.gov	Mozambique	Glaucoma and diabetic retinopathy	Fundus photography AI algorithms	Centro Hospitalar Universitário Lisboa Norte	2024	Not yet recruiting	100	No	NA
11	PACTR202502499779176	PACTR	Kenya	Primary care diagnosis	Large Language Model clinical decision support	PATH International	2025	Not yet recruiting	9000	No	NA

^aWhere a trial has been completed, the actual and planned enrolment figures are reported where available; for ongoing or not-yet-recruiting trials, the planned target is reported; NA = Manuscript not available.

Table 1: Trial Characteristics of the included trials (N = 11) in the network analysis.

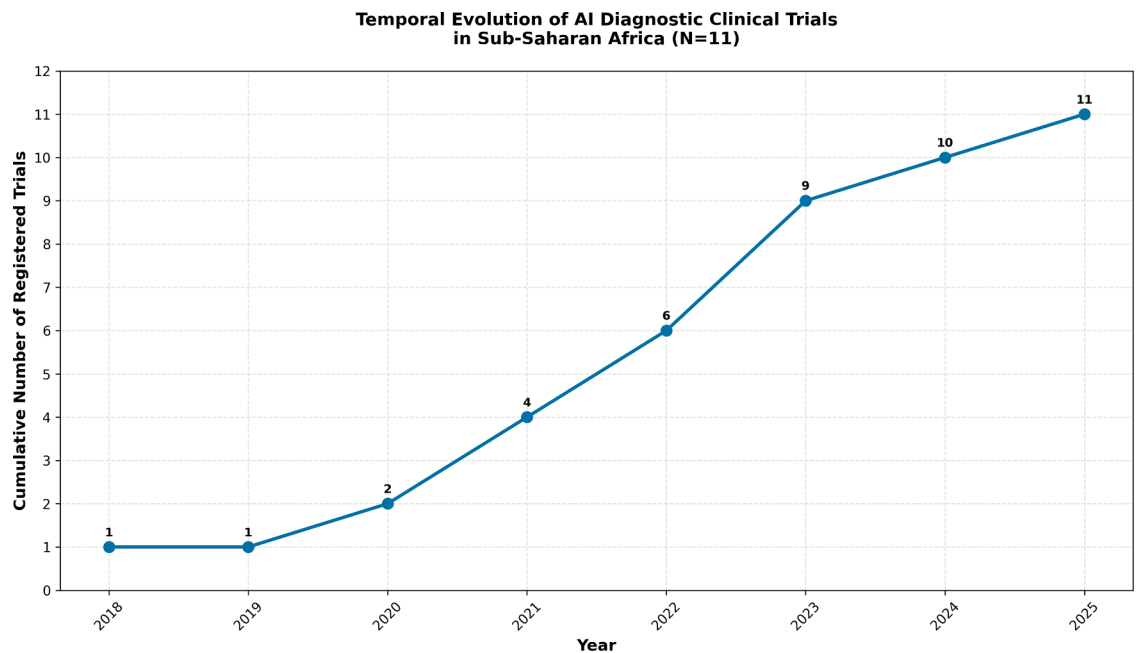


Fig. 1: Temporal evolution of registered AI diagnostic clinical trials in sub-Saharan Africa, 2018–2025 (N = 11). The y-axis shows the cumulative number of registered trials by end of each calendar year. The steep increase post-2020 reflects the nascent state of this research domain, coinciding with increased global attention to AI applications in health care and expanded funding for digital health innovations in resource-limited settings.

in which trials directly connect participating institutions, but institutions do not occupy positions bridging different research efforts.

Institutional centrality and network roles

Degree centrality values ranged from 0.022 to 0.178 (mean: 0.094 ± 0.051 , median: 0.089), identifying

Network Metric	Value	Description
Total nodes	46	11 trials + 35 institutions
Total edges	97	Direct trial-institution relationships (35 direct + 62 co-participation)
Network density	0.094	9.4% of possible connections present
Number of components	11	Completely disconnected trial-specific networks
Average degree centrality	4.22	Mean connectivity across all nodes
Degree Centralization	0.002	Near-zero centralization confirming absence of dominant hub nodes
Degree centrality (mean $\pm$ SD)	0.094 ± 0.051	Average and spread of individual node connectivity across the network
Degree centrality (median)	0.089	Typical node's average spread of individual node connectivity
Degree centrality (range)	[0.022, 0.178]	Minimum to maximum degree centrality values observed across nodes
Betweenness centrality (all nodes)	0	No institutional bridging positions
Closeness centrality (mean $\pm$ SD)	0.094 ± 0.051	Average accessibility between nodes
Closeness centrality (median)	0.089	Typical node's average shortest-path accessibility to all other nodes
Closeness centrality (range)	[0.022, 0.178]	Minimum to maximum closeness centrality observed across all nodes
Degree-closeness correlation (Spearman)	$p = 1.000$	Perfect correlation; fragmented structure
Degree-betweenness correlation	Undefined	Undefined due to betweenness constant at 0
Academic institutions	20	Largest institutional category
Funding organizations	10	Second largest category
Industry partners	3	Industry participation
Government entities	2	Ministry of Health representation

Table 2: Network characteristics of the constructed network of AI Diagnostic Trials in sub-Saharan Africa.

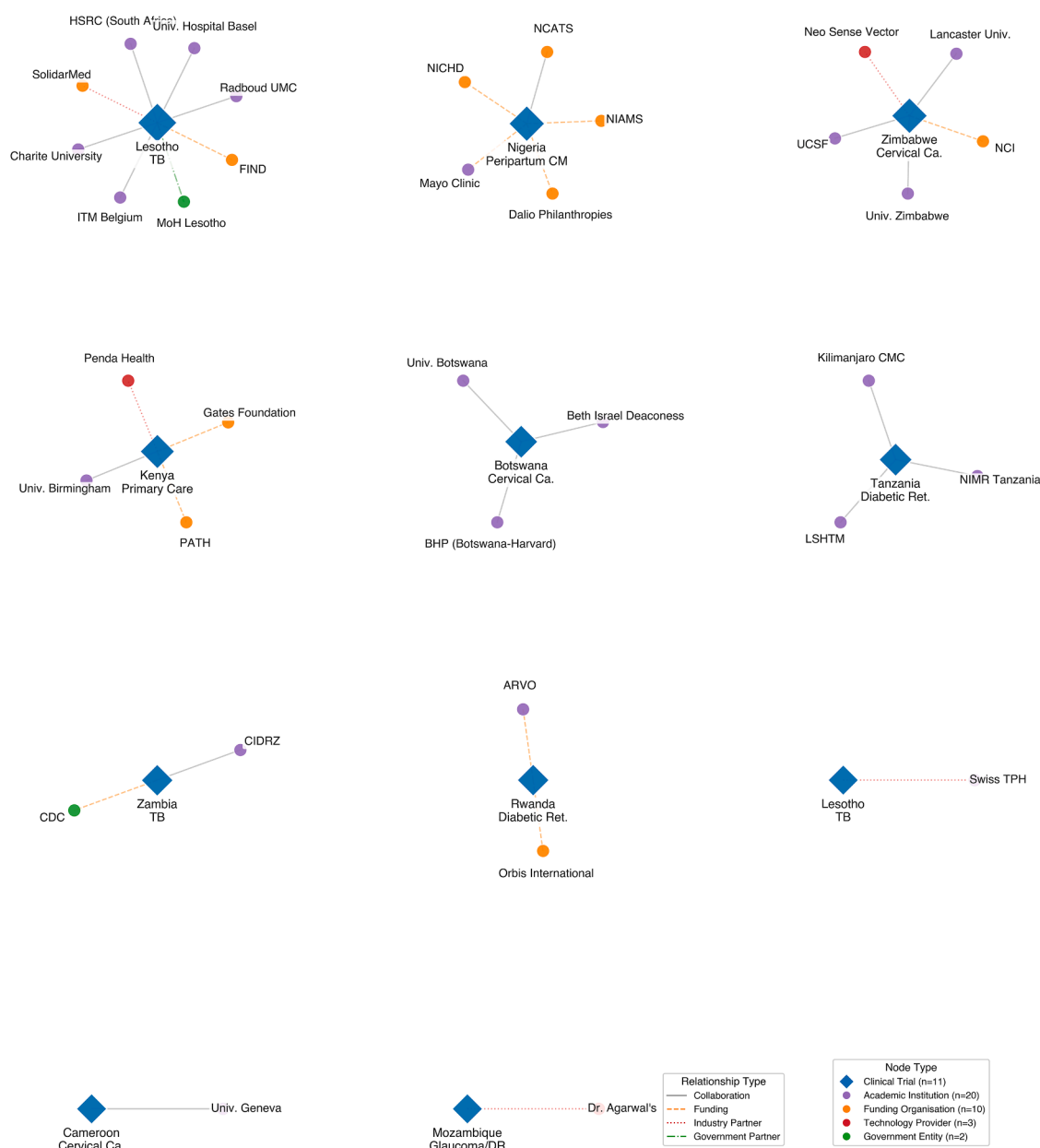


Fig. 2: Complete network of AI diagnostic clinical trials in sub-Saharan Africa (46 nodes, 97 edges; 11 trials, 35 institutions). Node shapes and colours represent entity type: clinical trials (blue diamonds, $n = 11$), academic institutions (blue circles, $n = 20$), funding organisations (orange circles, $n = 10$), technology providers (red circles, $n = 3$), and government entities (green circles, $n = 2$). Edge styles represent relationship type: direct trial-institution collaboration (solid lines), funding relationships (dashed lines), industry partnerships (dotted lines), and government partnerships (dash-dot lines). The 11 disconnected components reflect complete structural fragmentation of the network, with no institutional nodes occupying bridging positions across trials.

institutions with the most extensive collaborative networks within individual trials. Eight institutions achieved the highest degree centrality (0.178), all participating in a single multi-partner tuberculosis diagnostic trial: Charite University (Germany),

University Hospital Basel (Switzerland), SolidarMed, Radboud University Medical Center (Netherlands), Ministry of Health Lesotho, Institute of Tropical Medicine (Belgium), Foundation for Innovative New Diagnostics (Switzerland), and Human Sciences Research

Council (South Africa). Two institutions—Mayo Clinic and Lancaster University—demonstrated the next highest connectivity (degree centrality: 0.111), each participating in trials with moderate numbers of partners.

Among the 35 institutional nodes, academic institutions represented the largest category (19, 54.3%), followed by funding organizations (11, 31.4%), technology companies (4, 11.4%), and government entities (1, 2.9%). However, this sectoral diversity operated within trial-specific silos rather than facilitating cross-trial collaboration or knowledge transfer.

Absence of institutional bridging positions

Betweenness centrality analysis revealed complete absence of institutional bridging positions across the network. All nodes exhibited betweenness centrality of 0.0000 (median: 0.0000, range: [0.0000, 0.0000]), indicating that no institutions occupied structural positions facilitating information flow between different trials. This uniformly zero betweenness distinguishes between institutional participation (measured by degree centrality) and brokerage capacity (measured by betweenness centrality). While several institutions demonstrated high within-trial connectivity, none served as bridges enabling knowledge transfer, methodological diffusion, or collaborative learning across separate research efforts.

Closeness centrality values (mean: 0.094 ± 0.051 , median: 0.089, range: [0.022, 0.178]) exactly mirrored degree centrality distributions, confirming that in this fragmented network structure, accessibility between nodes is determined entirely by local connectivity within trial-specific components rather than by position within a larger integrated network.

The correlation between degree and closeness centrality was perfect (Spearman $\rho = 1.000$, $p < 0.001$), while correlations involving betweenness centrality could not be calculated due to zero variance across all nodes (Supplementary Fig. S1 and Table S3). This pattern confirms that in networks with multiple disconnected components, degree and closeness centrality convey identical information about node prominence, while betweenness centrality identifies the absence of inter-component bridging.

Discussion

This exploratory network analysis reveals the structural characteristics of AI diagnostic clinical trial networks in sub-Saharan Africa, identifying patterns that may influence ecosystem development and knowledge translation. We identified 11 registered clinical trials across 10 countries involving 35 institutions, with substantial temporal concentration (72.7% registered post-2020), reflecting the nascent state of this research domain. The network exhibited complete fragmentation into 11

disconnected components, with uniformly zero betweenness centrality indicating absence of institutional bridging positions between trials. While several institutions demonstrated high connectivity within individual trials, none occupied structural positions facilitating knowledge flow across separate research efforts.

Trial-centric network structure and knowledge transfer barriers

The observed network structure reveals a trial-centric organizational pattern with significant implications for ecosystem development and capacity building. In networks with bridging positions, institutions connecting otherwise separate clusters facilitate knowledge spillovers, cross-pollination of ideas, and transfer of methodological innovations between research domains.^{12,16} The absence of such positions in our network suggests that knowledge accumulated through one trial, such as regulatory navigation strategies, community engagement approaches, AI validation methodologies, data governance frameworks, does not necessarily flow to inform other trials through documented institutional networks.

This structural isolation creates missed opportunities for cross-trial learning. For instance, institutions participating in tuberculosis diagnostic trials and those involved in cervical cancer screening trials operate in separate network components despite facing similar AI implementation challenges, regulatory pathways, and community engagement needs in sub-Saharan African contexts. Lessons about engaging hard-to-reach populations from TB trials might inform recruitment strategies for cancer screening; AI model validation approaches could transfer between disease domains; regulatory approval experiences could guide subsequent trials. The zero betweenness structure demonstrates that such cross-trial learning is not occurring through institutional collaboration networks captured in trial registries.

Several dynamics may explain this structural pattern. First, the nascent field development reflected in temporal concentration (72.7% of trials registered post-2020) suggests insufficient time has elapsed for sustained cross-trial relationships and learning networks to form. Second, disease-specific research siloing may occur where tuberculosis researchers and cancer screening investigators operate in separate epistemic communities even within the same institution, limiting cross-domain knowledge transfer. Third, project-based funding structures create temporary partnerships that form around specific trials and dissolve after completion, rather than building sustained multi-trial relationships that accumulate and transfer knowledge. Fourth, absence of intentional bridge-building mechanisms, such as communities of practice spanning multiple trials, researcher rotation programs, or shared

methodological platforms, means institutional participation in multiple trials does not automatically translate into knowledge bridges.

Comparison to mature innovation ecosystems

In established research ecosystems, institutions occupying bridging positions serve critical functions: they facilitate knowledge diffusion, reduce network fragmentation, create resilience through redundant pathways, and accelerate innovation through cross-domain fertilization.^{17,18} The AI diagnostic trial network's lack of such bridges suggests it is in early formation stage with isolated rather than integrated innovation activities.

However, structural connectivity alone does not determine ecosystem functionality or sustainability. The innovation ecosystem identifies essential elements including diversity of actors, strength of connections, and presence of enabling assets such as funding, infrastructure, and supportive policies. Our network analysis reveals patterns in two dimensions, actor participation and structural connectivity, while acknowledging that functional ecosystem strengthening requires additional elements we cannot assess through trial registry data: relationship quality, equity of participation, adequacy of enabling policies and infrastructure, and mechanisms for knowledge retention and local ownership.

True ecosystem maturity and sustainability require functional outcomes beyond structural connectivity, including regulatory approvals demonstrating translation into practice, evidence of local research leadership rather than dependence on external partners, scale-up beyond trial settings into routine health system operations, and integration of validated technologies into national health policies and procurement systems. These functional indicators of sustainable innovation capacity cannot be inferred from network structure alone.

Power asymmetries and knowledge sovereignty

The network structure raises critical questions about power dynamics and knowledge sovereignty that warrant deeper investigation than registry data permit. While our analysis identifies which institutions participate and how they connect, it cannot determine governance arrangements, decision-making authority, data ownership, or intellectual property rights, dimensions essential for assessing whether partnerships foster equitable capacity development or reproduce colonial research relationships.

The concentration of high-degree institutions represents both opportunity and potential constraint. Hub institutions with extensive within-trial connectivity may indicate genuine regional research leadership and capacity concentration in Kenya, Nigeria, and South Africa. However, hub status could alternatively reflect

preferential funder relationships, historical path dependencies, or geographical proximity to donor offices that concentrate resources in established research centers. Without data on governance structures, our network analysis cannot distinguish between these interpretations.

Several power-related questions emerge from this structural analysis that future research should continue examining within decolonial and post-colonial theories of critical science.¹⁹ First, governance and decision-making: Do African institutions exercise meaningful control over trial design, research questions, and methodological choices, or do they primarily serve as implementation sites for externally-driven innovation agendas? Second, data ownership and intellectual property is now critical to data-intensive AI development and deployment: Who owns the datasets generated through these trials, and how are intellectual property rights distributed among partners?²⁰ Third, funding dependencies: How do resource flows from Global North donors shape research priorities, and what mechanisms exist to ensure African institutions can pursue locally-identified innovation needs?²¹ Fourth, capacity retention: Do observed partnerships build lasting research capacity and infrastructure that persists after trial completion, or do they create temporary capabilities that dissolve when external funding ends?²² Fifth, benefit distribution: How are benefits from successful AI diagnostic innovations, including revenues, recognition, and decision-making influence, shared among partners?²³

The absence of institutional bridging positions may itself reflect power asymmetries. If Global North institutions control funding and set research agendas, they may prefer bilateral partnerships with individual African institutions (creating the observed star-like trial structures) rather than supporting African inter-institutional collaborations that might shift agenda-setting power toward regional networks. Alternatively, competition among African institutions for limited research funding may discourage collaboration. Our structural findings cannot adjudicate between these mechanisms but highlight the need for research examining how power operates within these partnership networks.

Limitations of network metrics for assessing equity and sustainability

Network metrics characterize observed relational structure but have important limitations for assessing partnership equity or predicting sustainability. Structural connectivity does not indicate relationship quality. For example, two institutions connected in our network might have equitable co-investigator relationships or markedly asymmetric principal investigator/subcontractor arrangements. Registry data reveal institutional participation but not whether African partners shape

research questions, control data, author publications, or retain capacity after trials complete.

The trial-centric structure we observe may reflect rational specialization in an emerging field rather than problematic fragmentation. Early-stage ecosystems often exhibit exploration and experimentation across multiple parallel efforts before convergence on successful models.¹⁵ However, this structure could also indicate missed opportunities for knowledge accumulation and methodological refinement that characterize mature innovation ecosystems. Distinguishing between these interpretations requires longitudinal analysis tracking whether current partnerships evolve into sustained collaborations and whether knowledge circulates effectively despite structural fragmentation.

Our cross-sectional analysis characterizes current network structure but does not model temporal dynamics such as evolving edges, changing centrality over time, or network integration patterns. Dynamic network modeling was not appropriate for this nascent field for three reasons. First, with only 11 trials across seven years (2019–2025), insufficient data points exist per time period for robust temporal modeling (most years contain one or two trials, precluding meaningful statistical analysis of year-over-year changes). Second, 72.7% of trials registered in 2021 or later means insufficient time depth for observing partnership evolution, institutional learning, or network maturation processes that unfold over longer periods. Third, with 11 disconnected components and uniformly zero betweenness centrality, there are no evolving inter-trial connections to model, each trial forms an isolated network that does not interact with others over time. While we examined cumulative trial growth and temporal patterns in geographic distribution, disease focus, and funding sources, true dynamic network analysis requires larger samples and longer observation windows than currently available. Future research should employ longitudinal designs as this field matures.

Our analysis focuses on organizational-level networks documented in trial registries, excluding other ecosystem actors including patients and communities, regulatory agencies, ministries of health, data infrastructure providers, professional societies, and civil society organizations. This captures research relationships between institutions but not the full translation pathway from innovation to health system adoption. Network analysis of trial registries reveals which organizations participate in formal research partnerships but cannot assess whether regulatory capacity exists to approve AI tools, whether health ministries prioritize AI diagnostics in implementation plans, or whether communities trust and accept AI health technologies. Comprehensive ecosystem mapping would require stakeholder interviews, policy analysis, and community engagement assessment beyond trial registry data.

Critically, however, this boundary is itself substantively meaningful rather than merely a methodological constraint. Clinical trial registration is not universally mandated in sub-Saharan Africa: while ICMJE-affiliated journals require prospective registration, national regulatory frameworks vary considerably across the region, and the PACTR, the continent's only WHO-endorsed primary registry, operates on a voluntary basis for most member states. Industry deployments, government-commissioned pilots, and commercially funded evaluations of AI diagnostic tools almost certainly constitute a substantial and growing share of actual AI diagnostic activity in the region yet operate largely outside formal registry systems and are therefore absent from our network. Africa accounts for under 3% of global clinical trial registrations despite carrying approximately 20% of the global disease burden, and among completed registered trials only 16.5% had posted results on [ClinicalTrials.gov](https://clinicaltrials.gov), indicating substantial underreporting even within the registered landscape.²³

Our $N = 11$ therefore almost certainly represents a cross-section of a larger, partially invisible landscape of activity. The registered network is, however, the only analytically tractable unit for social network analysis: informal deployments leave no documented institutional relationships to map. The sparseness and fragmentation we observe should accordingly be read not as a sampling artefact but as a substantive reflection of the field's current governance and transparency architecture—a finding that is itself informative about ecosystem maturity, not merely a limitation of our method.

Our finding of zero betweenness centrality identifies absence of documented inter-trial institutional bridges but cannot determine whether informal knowledge transfer occurs through other mechanisms such as investigator mobility, conference participation, or shared training programs. Network analysis of trial registries captures formal organizational relationships but may underestimate knowledge flows operating through interpersonal networks or communities of practice.

Intervention opportunities for ecosystem strengthening

The structural patterns we identify suggest several intervention opportunities, recognizing that ecosystem strengthening requires coordinated action across multiple dimensions beyond network structure alone.⁸ First, cross-trial learning communities could create forums where investigators from different trials share methodological innovations, regulatory experiences, and implementation lessons, building bridges that formal institutional partnerships have not created. Second, researcher mobility programs could facilitate movement of technical staff and investigators between

trials to transfer tacit knowledge and build relationships. Third, shared platforms for common challenges, including AI validation frameworks, data governance templates, and community engagement toolkits, could be developed collaboratively and refined across trials. Fourth, multi-phase consortia where funding supports institutions conducting sequential or parallel trials that intentionally build on each other could transform isolated projects into accumulating knowledge systems.

However, these structural interventions alone are insufficient without attention to power dynamics and knowledge sovereignty. Ecosystem strengthening must ensure African institutions exercise genuine governance over research agendas, retain intellectual property rights, build permanent rather than temporary capacity, and participate equitably in benefits from successful innovations. This requires funding mechanisms that support African-led consortia, policies requiring equitable authorship and data ownership, and deliberate capacity retention strategies that transcend individual trial lifecycles.

Strengths and limitations

This study's primary strength is comprehensive mapping of all registered AI diagnostic clinical trials in sub-Saharan Africa, providing a complete census rather than a sample. We applied established network analysis methods to characterize ecosystem structure, following precedents in health systems research.^{10,11,24} The exploratory case study approach appropriately matches the nascent state of this field, generating hypotheses about ecosystem development patterns rather than testing predetermined theories.

However, several limitations constrain interpretation. The small absolute number of trials ($N = 11$), while representing the complete population, limits statistical inference and makes network metrics sensitive to individual trial characteristics. Our analysis captures only formally documented relationships through trial registries and published reports; informal collaborations, shared training programs, or investigator mobility operating outside registered trials is thus not captured. We also cannot assess relationship quality, power dynamics, or knowledge transfer effectiveness from structural connectivity alone—critical dimensions for understanding whether observed patterns foster equitable capacity building.

The cross-sectional design precludes assessment of ecosystem evolution; longitudinal analysis tracking how partnerships form, strengthen, or dissolve over time would provide insights into network dynamics that single-timepoint analysis cannot reveal. Our focus on clinical trials excludes other innovation ecosystem components including pilot demonstrations, implementation science studies, and government technology deployments that may connect institutions outside formal research partnerships. Finally, our geographic scope

encompasses diverse sub-Saharan African contexts but cannot account for country-specific variations in regulatory environments, funding landscapes, or research infrastructure that shape local innovation ecosystems.

Implications for policy and practice

For policymakers and funders, these findings suggest that building sustainable AI diagnostic innovation capacity in sub-Saharan Africa requires more than funding individual trials. Deliberate ecosystem strengthening interventions, creating cross-trial learning networks, supporting researcher mobility, developing shared platforms, and structuring multi-phase consortia, may accelerate knowledge accumulation and capacity development. However, structural interventions alone are insufficient; funders must simultaneously address power asymmetries through governance requirements ensuring African institutional leadership, equitable intellectual property arrangements, and capacity retention mechanisms.

For research institutions, the trial-centric structure highlights opportunities to build strategic partnerships that transcend individual projects. Institutions participating in AI diagnostic research might proactively create communities of practice, share methodological resources, or develop joint training programs even absent formal funding for such collaboration. African research networks could advocate for funding mechanisms supporting regional consortia rather than isolated bilateral partnerships with Global North institutions.

For the global health research community, our findings underscore the distinction between connectivity and integration in innovation ecosystems. Current partnership patterns create participation opportunities for African institutions but may not be building the relational infrastructure required for knowledge accumulation, methodological refinement, and capacity retention that characterize mature, sustainable innovation ecosystems. Moving from isolated trials to integrated ecosystems requires intentional attention to structural connectivity, power dynamics, and functional sustainability dimensions simultaneously.

Conclusion

This exploratory network analysis reveals that AI diagnostic clinical trial networks in sub-Saharan Africa exhibit a trial-centric structure with complete fragmentation and absence of institutional bridging positions. While these structural patterns reflect the nascent state of the field, they suggest missed opportunities for knowledge transfer and capacity building across trials. However, structural connectivity alone cannot determine ecosystem health, equity, or sustainability, dimensions requiring examination of governance arrangements, power dynamics, and functional outcomes beyond what network analysis of trial registries can reveal.

Future research should employ mixed methods combining network analysis with qualitative investigation of partnership governance, power dynamics, and knowledge transfer mechanisms; longitudinal designs tracking ecosystem evolution and capacity retention; and examination of how observed network structures relate to functional outcomes including regulatory approvals, local ownership, and health system integration. Understanding whether current partnership patterns foster equitable, sustainable innovation capacity or reproduce dependency relationships requires moving beyond structural description to examine the quality, equity, and functionality of relationships within these emerging networks.

Contributors

JIF and HR conceived and designed the study, with input from JBN. JIF and MT conducted the systematic registry searches, extracted data, and performed network analyses. JIF and HR wrote the first draft of the manuscript. EJM provided senior supervision. All authors critically revised the manuscript for important intellectual content and approved the final version for submission. JIF and MT directly accessed and verified the underlying data. All authors had full access to all data in the study and accept responsibility for the integrity of the data and the accuracy of the analyses. JIF had responsibility for the decision to submit the manuscript for publication.

Data sharing statement

The data underlying this analysis are derived from publicly accessible clinical trial registries (ClinicalTrials.gov, PACTR, and the WHO International Clinical Trials Registry Platform). The extracted network dataset, including institutional nodes, edges, and centrality metrics, is available in Microsoft Excel format and will be made available upon reasonable request to the corresponding author (james.forrest@ubc.ca) beginning at the time of publication. No study protocol, statistical analysis plan, or individual participant data exist, as this study used only publicly registered trial information. Data may be shared for non-commercial research purposes with a completed data access request submitted to the corresponding author.

Declaration of interests

We declare no competing interests.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.lanafr.2026.100064>.

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